



STRULE AUTOMATION

Industrial Controls · Bay Area & Northern California

DIAGNOSTIC FIELD GUIDE

Is It the Sensor or the Process?

Who this is for and what it does: maintenance techs, plant engineers, and controls people staring at a reading that looks wrong. This guide helps you decide whether the instrument is lying or the process really is doing what the number says, before you start changing setpoints or swapping parts.

How to use this

Work top to bottom. Find the symptom that matches what you're seeing, run the checks in order, and write down what you find as you go. The cheap checks come first on purpose. Most of the time you can settle the sensor-versus-process question without a single spare part, just by making the reading disagree or agree with a second source of truth.

One habit worth building: before you touch anything, note the current reading, the time, and what the process is actually doing (running, idle, filling, heating). Half of these calls get solved by comparing the number against something you already know to be true.

The core question

A reading is a claim: the level is 40%, the flow is 12 gallons per minute (GPM), the temperature is 340 degrees. Your job is to find one independent thing that agrees or disagrees with that claim. When two sources that should move together don't, one of them is wrong, and now you have a real problem to chase instead of a hunch.

Independent means the second source does not share the failure. A second transmitter on the same power supply, cable tray, and ground is not fully independent. A tape measure, a sight glass, the motor's current draw, a delivery ticket, or a hand-held meter usually is.

Symptom → likely causes → what to check and log

Reading is frozen or barely moves while the process is clearly changing

Likely cause	What to check	What to log
Sensor fouled, coated, or plugged (level, flow, analytical)	Pull or isolate and look at the wetted end. Coating, scale, product buildup?	Condition of the sensing element, last clean date
Impulse line or tap plugged (pressure, differential-pressure level/flow). The impulse line is the small-bore tubing that connects the process to a pressure or differential-pressure transmitter	Crack the vent/drain, check for free flow. Compare against a direct read if one exists	Whether line cleared, any debris
Signal railed or saturated	Read the raw milliamp value at the transmitter and again at the input card	mA at both ends, engineering value shown
Controller in manual, or the logic is holding the value	Check the mode and any latch that is freezing the reading: a "first-out" capture (the logic grabbed and held the first value that tripped) or a "seal-in" (a rung that keeps holding itself on after the trigger clears)	Mode, active holds

Reading is noisy, jumpy, or bounces around a plausible average

Likely cause	What to check	What to log
Electrical noise coupled into the loop	Note whether the jumps line up with a nearby drive, contactor, or pump starting	Time-correlate noise with nearby equipment
Shield or ground problem	Verify shield is landed at one end only, and that the drain wire isn't floating or doubled	Shield scheme, where it's landed
Real process turbulence	Look at the process. Splashing, boiling, two-phase flow, agitator wash near a probe?	What the process is physically doing
Loose termination or corroded connection	Wiggle-test terminations (power down where required), inspect for green/black on lugs	Any connection that changed the reading when moved

Reading is smooth and believable but wrong by a consistent amount

Likely cause	What to check	What to log
Calibration drift, zero or span off	Compare against a known reference at two points if you can (not just one)	Reference value vs indicated at low and high
Zero shift from mounting, static head, or temperature	Check zero at a known condition (empty tank, no flow, ambient)	Indicated value at the known-zero condition
Wrong scaling or range in the transmitter or the input card	Confirm the range in the device matches the range in the logic	Device range, card scaling, engineering units
Sensor aged or partially fouled (offset, not total block)	Clean and re-read. Did the offset shrink?	Before/after clean readings

Reading looks fine but the process behaves wrong (or vice versa)

This is the important one, and it's where the sensor-versus-process question actually lives. It's also the one that fools me most often, because a steady, believable number is the last thing you think to doubt. The number says 40% level and holding, but the pump is short-cycling and the outlet is spitting air. Or the number says the tank is full and overflowing, but nothing is coming out the vent. Here you stop trusting the instrument and go find a second witness.

Cross-check	How to use it	Sensor is suspect when...
Motor current or power	A pump moving real product draws real current. A conveyor carrying real load draws real load. Know your pump type, though: a centrifugal pump dead-headed against a shut valve draws <i>less</i> current, while a positive-displacement pump against a closed discharge spikes current and pressure. So use current as a "moving or not" clue, not a flow gauge	Current and reading tell opposite stories once you account for pump type
A second, independent sensor	Compare against a different measurement principle if available (e.g., a radar level vs a pressure-based level)	Two independent principles disagree beyond their combined error
Manual/physical check	Tape a tank, use a sight glass, catch a bucket and time it, feel a pipe for heat	Hand check and instrument disagree materially

Cross-check	How to use it	Sensor is suspect when...
Mass/volume balance	What went in should show up as level or weight. Compare a delivery ticket or batch total to the tank move	The books don't close and only one instrument explains the gap

Signal-integrity basics (the 4-20 mA reality check)

Most process loops still run on a 4-20 milliamp (mA) current signal, and current loops are stubborn in a useful way: the same current flows everywhere in the loop, so you can read it at any break point and it should match.

- **Read the loop current, not just the engineering value.** A series reading at the transmitter and again at the input terminals should agree. Doing that means breaking the loop in series or using a true DC milliamp clamp or loop calibrator — an ordinary AC clamp meter won't read a DC loop, and the value on a HART communicator or the transmitter's own display is a reported number, not the measured analog loop current. If the field device says 12 mA and the card reads something else, you have a wiring, termination, or card problem, not a process problem.
- **A signal driven hard toward zero or pinned high is a fault, not a reading.** Healthy transmitters legitimately live down to about 3.8 mA and up to about 20.5 mA at range extremes, so don't call a valid 3.9 mA a failure. The recognized fault bands (NAMUR NE43) sit at or below roughly 3.6 mA and at or above roughly 21 mA. The narrow sliver between the live range and the fault band — 3.6 to 3.8 mA at the bottom, 20.5 to 21 mA at the top — isn't a contradiction; it's deliberate under-range and over-range margin, a guard band so normal noise at the extremes doesn't flip a healthy loop into a fault state. A signal driven down near 3.6 mA is usually a broken wire, wet terminal, or dying transmitter; pinned near 21 mA is often a saturated or failed device. Many input cards can flag these as under-range, over-range, or open-loop faults if you enable it, and it's worth enabling. Older or basic cards may not offer those flags at all; on those the raw milliamp value is your fault detector — a signal parked at 3.6 or 21 mA reads the same whether the card annunciates it or not.
- **Grounding and shielding.** Instrument cable shields should be landed at one end only, normally the control-panel end, and left isolated at the field end. Landing both ends invites a ground loop, where a small voltage difference between two grounds pushes current through the shield and couples noise into your signal. A shield landed at neither end does nothing. I've spent most of a shift chasing a noisy level reading that turned out to be a shield landed at both ends, so if you find one bonded at both ends, that's a common source of the "noisy for no reason" complaint.
- **Cable routing near drives.** Variable-frequency drives (VFDs) are loud, electrically. Signal cable run in the same tray as drive output cabling, or crossing it in parallel for a long stretch, will pick up noise. Separation and crossing at right angles help. If the noise appears only when a specific drive is running, you've found your coupling path.

- **One power supply feeding many loops.** A sagging or noisy supply shows up on several instruments at once. If three unrelated readings got jumpy on the same day, suspect something common: power, a ground, a shared cable route.

A quick decision sequence

1. **Start here:** Does the reading make physical sense right now? If the tank can't be at 90% because you just drained it, the instrument is the problem. Skip to the instrument checks.
2. **If it's plausible but suspicious,** find one independent cross-check (current, second sensor, manual). Agreement points at the process. Disagreement points at the instrument.
3. **If they disagree,** read the raw loop current at both ends. Mismatch between ends is wiring or card. Match between ends but wrong value is the sensor or its calibration.
4. **If the loop is electrically healthy,** check calibration at two points and inspect the wetted end for fouling. Most "wrong but steady" readings die here.
5. **If it's healthy, calibrated, and clean, believe it.** The process really is doing what the number says, and now you have a process problem to solve. That's a good outcome. You stopped chasing the instrument.

10-minute triage (run this before calling anyone)

- ☐ Write down the current reading, the time, and what the process is physically doing.
- ☐ Find one independent cross-check (motor current, second sensor, tape/sight glass, a bucket and a stopwatch) and compare.
- ☐ Read the raw 4-20 mA loop current at the transmitter and at the input card; confirm they agree and sit inside the healthy live range, not down near 3.6 mA or up near 21 mA.
- ☐ Confirm the controller loop is in the mode you think it's in (auto vs manual) and no hold or latch is freezing the value.
- ☐ Eyeball the field wiring: loose lugs, corrosion, water in the enclosure, shield landed at both ends.
- ☐ Note whether noise or jumps line up with a nearby drive, pump, or contactor cycling.
- ☐ If safe and permitted, inspect or isolate the wetted end for coating, scale, or a plugged tap.
- ☐ Check the device range against the logic scaling for the same engineering units.

If you get through that list, you'll usually know which side of the fence the fault is on, and whoever you call next gets a real starting point instead of "the number looks funny."

Safety note

The trap specific to this work is deciding the sensor is lying and then forcing the value or strapping out the trip so the process keeps running — when the reading may be correct and the tank really is at that high level or the process really is over temperature. Don't defeat a protective interlock, bypass a

safety instrumented function, or force a value to make a reading "behave." Retuning a nuisance trip, adding a sensible filter, or fixing a grounding problem is legitimate engineering; a protective function that keeps tripping should be treated as data about a real condition until you've proven otherwise. Follow your site's lock-out/tag-out and energized-work rules for any check that opens an enclosure or breaks a live loop.

Honest limits

This guide narrows the question but doesn't close every case: an intermittent won't show in a 10-minute look (which is why logging beats any single reading), some wrong readings stay internally consistent until an independent principle or a mass balance over time catches them, and two sensors can agree while both are wrong if they share a fouling mechanism or a bad common reference — so use the cross-checks as evidence, not proof.

If this fault keeps getting handed around

When a reading has bounced between the instrument vendor, the drive vendor and the controls integrator and each says it's someone else's box, the fault usually lives in the seams between systems rather than inside any one of them. I work vendor-neutral, with no product line to defend, so I can look at the whole loop — grounding, signal path, calibration and logic. If you're stuck there, it's worth a conversation: struleautomation.com or info@struleautomation.com

— Jonathan Gilmour, Strule Automation